

STA 540L Case Study 1: Reproduction

Tracy Sun

2026-01-28

Table 1

Table 1: Summary of National Institute on Drug Abuse Clinical Trials Network Social Media Pre-exposure Prophylaxis Study, 2020, population sociodemographic and behavioral characteristics (N=254).

Characteristic	Value
Age in Years, median (IQR)	25 (21-27)
Ethnicity, n (%)	
Hispanic/Latinx	66 (26)
Race, n (%)	
American Indian or Alaskan Native	1 (0.4)
Black or African American	196 (78.4)
White	28 (11.2)
Other	14 (5.6)
Multiracial	11 (4.4)
History of PrEP uptake, n (%)	
Never taken PrEP	232 (91.3)
Have taken PrEP, but not within the past 6 months	22 (8.7)
Number of male sex partners in the past 90 days, median (IQR)	4 (3-6)
Condom use, n (%)	
Never	36 (14.2)
Sometimes	108 (42.5)
About half the time	37 (14.6)
Most of the time	68 (26.8)
Always	5 (2)
Condomless receptive anal sex in the past 90 days, n (%)	210 (82.7)
Ever tested for HIV during lifetime, n (%)	191 (75.2)
If tested for HIV, median (IQR)	
Months since last HIV test	10.9 (6.1-21.1)
If not tested for HIV, n (%)	63 (24.8)
Main reasons cited by the 63 participants for not getting tested, n (%)	
Unlikely to be exposed to HIV	8 (12.7)
Afraid of testing HIV-positive	26 (41.3)
Did not want to think about HIV/HIV-positive	8 (12.7)
Worried about names being reported if positive	3 (4.8)
Dislike for needles	5 (7.9)
Unable to trust that the results will be confidential	3 (4.8)
Unaware of where to get tested	7 (11.1)
Other reasons	3 (4.8)

Primary analysis

Fitting Poisson Model

Estimating cell-specific rates

Site Name	Wave	Number of days per wave	Platform Type	Rate	95% SE
Facebook	1	70	Social Media	0.19	0.05
Instagram	2	38	Social Media	0.34	0.09
Grindr	1	70	Dating Apps	0.13	0.04
Jack'd	2	38	Dating Apps	3.29	0.29
Google	1	70	Info Sites	0.24	0.06
Bing	2	38	Info Sites	0.00	0.00

Contrasts

Wave	Contrast	P-value
1	Grindr vs Google	0.3686
1	Grindr vs Facebook	0.4665
1	Google vs Facebook	0.4665
2	Jack'd vs Bing	0.0002
2	Jack'd vs Instagram	0.0000
2	Bing vs Instagram	0.0238

Secondary analysis

a) Tobacco, alcohol, prescription medications, and other substance use

Substance	Category	Ordered test kit	Percent	Did not order test kit	Percent	P-value
Alcohol	None	52	29.4	16	20.8	0.35
	Problem Use	46	26.0	21	27.3	
	High Risk Substance Use	79	44.6	40	51.9	
Cannabis	None	89	50.3	32	41.6	0.096
	Problem Use	37	20.9	12	15.6	
	High Risk Substance Use	51	28.8	33	42.9	
Stimulants	None	166	93.8	69	89.6	0.299
	Problem Use/High Risk Substance Use	11	6.2	8	10.4	
Opioid	None	173	97.7	75	97.4	1
	Problem Use/High Risk Substance Use	4	2.3	2	2.6	
Sedative	None	168	94.9	72	93.5	0.766
	Problem Use/High Risk Substance Use	9	5.1	5	6.5	
Prescribed Stimulant	None	172	97.2	74	96.1	0.702
	Problem Use/High Risk Substance Use	5	2.8	3	3.9	

b) Stage of Health Behavior Change

Attitude	Description	Ordered test kit	Percent	Did not order test kit	Percent	P-value
Precontemplation	I do not see any need to regularly test for HIV	7	4.0	2	2.6	0.251
Contemplation	I think I should get tested for HIV regularly, but I am not sure	31	17.5	8	10.4	
Determination	I am ready to start getting regularly tested for HIV	87	49.2	49	63.6	
Action	I am trying to get tested regularly for HIV	40	22.6	12	15.6	
Maintenance	I have been getting tested for HIV regularly over the past few years	12	6.8	6	7.8	

c) Attitudes toward human immunodeficiency virus (HIV) testing

Attitude	Response	Ordered test kit	Percent	Did not order test kit	Percent	P-value
Getting tested for HIV helps people feel better	Agree	169	95.5	70	90.9	0.16
	Disagree	8	4.5	7	9.1	
Getting tested for HIV helps people from getting HIV	Agree	148	83.6	62	81.6	0.717
	Disagree	29	16.4	14	18.4	
People in my life would leave if I had HIV	Agree	59	33.7	37	48.1	0.035
	Disagree	116	66.3	40	51.9	
People who tested positive for HIV should hide it from others	Agree	18	10.2	9	11.7	0.825
	Disagree	159	89.8	68	88.3	
I would rather not know if I have HIV	Agree	27	15.3	15	19.5	0.463
	Disagree	150	84.7	62	80.5	

d) Attitudes toward human immunodeficiency virus (HIV) treatment (continuous scale from 1 [strongly disagree] to 7 [strongly agree])

Statements	Ordered test kit: Mean (SD)	Did not order test kit: Mean (SD)	P-value
I am less threatened by the idea of being HIV positive than I used to be	4.1 (2)	3.8 (2.2)	0.421
I am less worried about HIV infection than I used to be	3.7 (2)	3.5 (1.8)	0.400
I think HIV/AIDS is less of a problem than it used to be	3.9 (2.1)	3.6 (2.1)	0.413
I think HIV/AIDS is a less serious threat than it used to be because of new HIV/AIDS treatments	4.2 (2)	3.8 (2.1)	0.225
I am much less concerned about becoming HIV positive myself because of new HIV/AIDS treatments	3.4 (2)	3 (1.9)	0.274
I think that condom use during sex is less necessary now that new HIV/AIDS treatments are available	2.4 (1.8)	2.2 (1.5)	0.971
I think that someone who is HIV positive now needs to care less about condom use	1.9 (1.7)	2.5 (2.1)	0.064
I think that the need for condom use is less than it used to be, because you can always start new treatments	2.5 (1.9)	2.5 (2)	0.767
I think that someone who is HIV positive and uses new HIV/AIDS treatments can be cured	3.2 (1.9)	3.6 (2.2)	0.199
I think that new HIV/AIDS treatments can eradicate the virus from your body	3.1 (1.9)	3.8 (2.1)	0.029

e) Human immunodeficiency virus (HIV)-related stigma among study participants

HIV stigma	Response	Ordered test kit	Percent	Did not order test kit	Percent	P-value
I feel afraid of people living with HIV/AIDS	Strongly agree	12	6.8	7	9.1	0.613
	Agree	9	5.1	2	2.6	
	Somewhat agree	21	11.9	9	11.7	
	Neither agree nor disagree	20	11.3	9	11.7	
	Somewhat disagree	11	6.2	6	7.8	
	Disagree	32	18.1	17	22.1	
I could not be friends with someone who has HIV/AIDS	Strongly disagree	72	40.7	27	35.1	0.032
	Strongly agree	1	0.6	4	5.2	
	Agree	1	0.6	0	0.0	
	Somewhat agree	3	1.7	1	1.3	
	Neither agree nor disagree	9	5.1	6	7.8	
	Somewhat disagree	7	4.0	1	1.3	
People who get HIV/AIDS through sex or drug use got what they deserve	Disagree	26	14.7	19	24.7	0.332
	Strongly disagree	130	73.4	46	59.7	
	Strongly agree	2	1.1	0	0.0	
	Agree	2	1.1	1	1.3	
	Somewhat agree	3	1.7	2	2.6	
	Neither agree nor disagree	8	4.5	7	9.1	
I feel anger toward people with HIV/AIDS	Somewhat disagree	6	3.4	2	2.6	0.213
	Disagree	24	13.6	12	15.6	
	Strongly disagree	132	74.6	53	68.8	
	Strongly agree	1	0.6	0	0.0	
	Agree	0	0.0	0	0.0	
	Somewhat agree	0	0.0	0	0.0	
	Neither agree nor disagree	11	6.2	5	6.5	
	Somewhat disagree	3	1.7	0	0.0	
	Disagree	19	10.7	16	20.8	
	Strongly disagree	143	80.8	56	72.7	

f) Medical mistrust

Medical mistrust	Response	Ordered test kit	Percent	Did not order test kit	Percent	P-value
You'd better be cautious when dealing with health care organizations	Strongly agree	37	20.9	17	22.4	0.503
	Agree	66	37.3	32	42.1	
	Disagree	42	23.7	14	18.4	
	Strongly disagree	32	18.1	13	17.1	
Patients have sometimes been deceived or misled by health care organizations	Strongly agree	32	18.2	11	14.3	0.413
	Agree	83	47.2	34	44.2	
	Disagree	33	18.8	22	28.6	
	Strongly disagree	28	15.9	10	13.0	
When health care organizations make mistakes they usually cover it up	Strongly agree	29	16.7	10	13.2	0.222
	Agree	83	47.7	29	38.2	
	Disagree	36	20.7	30	39.5	
	Strongly disagree	26	14.9	7	9.2	
Health care organizations have sometimes done harmful experiments on patients without their knowledge	Strongly agree	30	17.0	14	18.2	0.413
	Agree	73	41.5	25	32.5	
	Disagree	51	29.0	26	33.8	
	Strongly disagree	22	12.5	12	15.6	
Health care organizations don't always keep your information totally private	Strongly agree	36	20.6	11	14.5	0.371
	Agree	64	36.6	29	38.2	
	Disagree	47	26.9	23	30.3	
	Strongly disagree	28	16.0	13	17.1	
Sometimes I wonder if health care organizations really know what they are doing	Strongly agree	22	12.5	9	11.7	0.965
	Agree	66	37.5	28	36.4	
	Disagree	60	34.1	30	39.0	
	Strongly disagree	28	15.9	10	13.0	
Mistakes are common in health care organizations	Strongly agree	21	11.9	8	10.7	0.638
	Agree	89	50.6	36	48.0	
	Disagree	46	26.1	23	30.7	
	Strongly disagree	20	11.4	8	10.7	

Figure S1

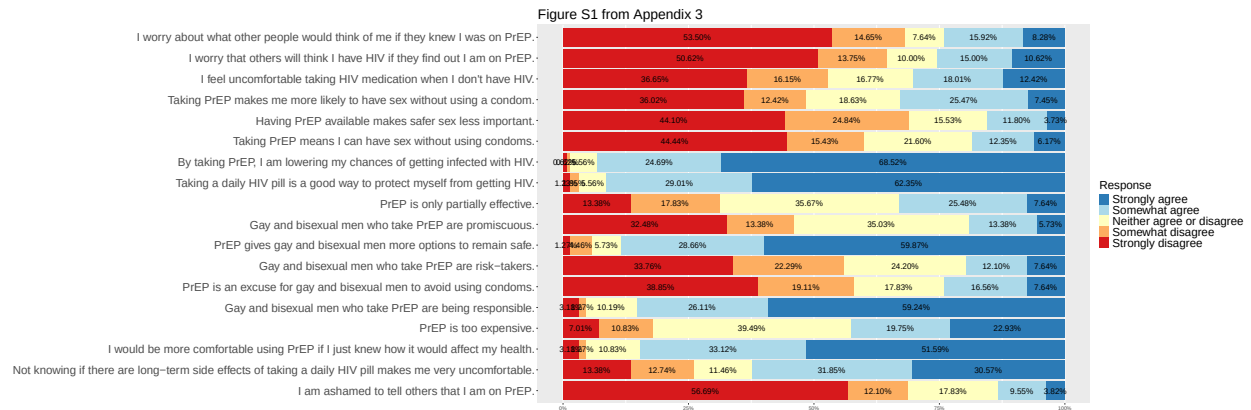


Figure S2

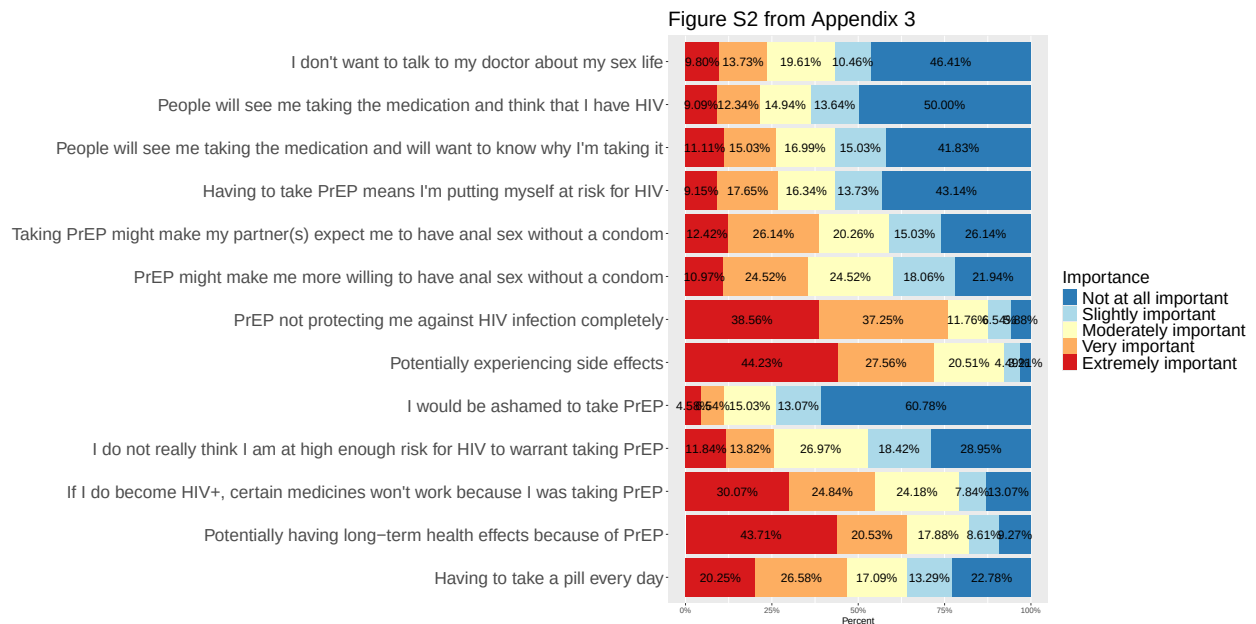
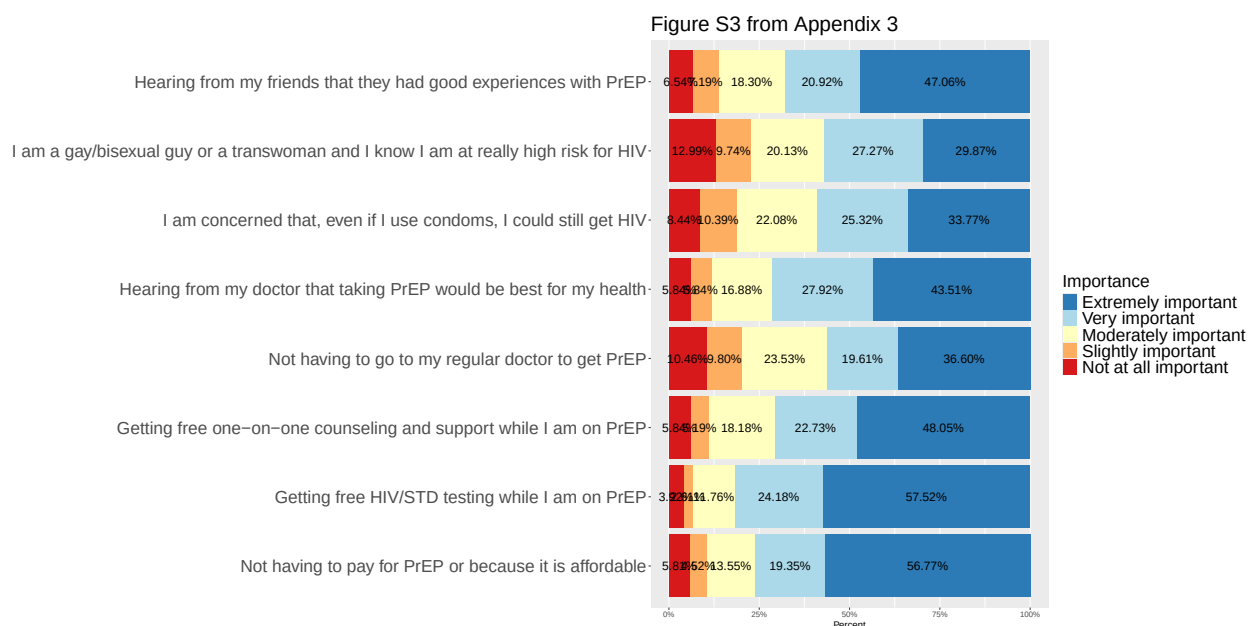


Figure S3



Reflection

Primary Analysis

Table 1 can be almost perfectly replicated, with a few small differences and notes:

- The race section's counts and percentages are out of 250 instead of 254, as there are 4 participants whose race was left as NA so they were excluded.
- There was one individual who took the HIV test but also marked a reason not to take it, so they were excluded from the "reasons not to take HIV test" section.
- The IQR for the age in years, as well as the median and IQR for months since last HIV test are slightly different from the original Table 1.

As for the contrast significance testing, the p-values obtained are slightly different from the original, although the conclusions on significance are the same, with all the platforms in Wave 1 not being significantly different from each other at the 0.05 level, and all the platforms in Wave 2 being significantly different from each other (mainly due to Jack'd having a lot more test kits ordered than any other platform). However, due to Bing having exactly zero test kits ordered, with the original model, the p-values for both of Wave 2's contrasts that involve Bing are highly non-significant. This may be due to slight differences in how R and SAS calculate the test statistics. However, by adding a small pseudocount of 0.5 to Bing's test kit number to stabilize the estimate, the p-values flip from being insignificant to a similar significance as the paper, with the contrasts for Bing vs Instagram being slightly less significant than Bing vs Jack'd or Jack'd vs Instagram.

Secondary Analysis

The secondary analysis, depicted in Appendix 3, is also able to be almost perfectly replicated. The only difference is that for the first table on alcohol/substance use, I was unable to find and account for the 5 rows for the sedative and prescribed stimulants category that were listed as missing, so my counts and p-values were slightly off.

References

Manuscript of the original paper and its appendices, which were followed to reproduce the results, can be found here: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9591705/>

<https://bookdown.org/yihui/rmarkdown-cookbook/text-width.html>

<https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/wilcox.test>

ChatGPT was used to generate some sections of this code as well as for some editing and debugging.